Euribor macro strategy: €50,000 account backtest

7 October 2026

1. Hypothesis

The frozen specification earns a positive return on the 2nd quarterly Euribor contract. It has two parts. **ECB rate momentum** sets the default position: long if the deposit facility rate was cut in the last three months, short if it was raised, otherwise the position is kept. **The risk factors** can override it: when the risk-factor score is above +0.5 the position is long, when it is below −0.5 the position is short, whatever the ECB did last. The position is decided at each month-end and held for the next month. This test asks what that edge looks like as a real account: €50,000 of capital, whole contracts, euro costs, margin, and a sizing rule, since the specification itself only says long or short.

The risk factors. Three groups of monthly macro series, each turned into a score and averaged into one number:

Group	What it captures	Series (sign: + argues for a higher futures price, i.e. lower rates)	Publication lag
Rates and policy	where policy and the money-market curve have been moving	deposit rate change over 3 and 6 months (−); 12M minus 3M Euribor slope (+) and its 3-month change (−); 12M Euribor 3-month change (−); 3M Euribor minus the deposit rate (+); 2nd-contract price momentum over 3 and 12 months (+)	none
Inflation	whether the ECB has room to ease	headline and core HICP, year on year (−), and their 3-month changes (−); unemployment rate, 6-month change (+)	HICP 1 month, unemployment 2 months
Sentiment and stress	financial and political stress that pushes the ECB toward easing	Europe economic policy uncertainty index, level and 3-month change (+); Italy plus Spain minus Germany plus France uncertainty, level and change (+); Italian and Spanish minus German 10-year yield spread, level and change (+); ECB composite systemic stress indicator, level and change (+); economic sentiment indicator, level and change (−)	uncertainty and sentiment 1 month, others none

Each series is converted to an expanding z-score (at least 36 months of history), clipped to ±3 and multiplied by its sign; the signs were fixed from theory, not fitted. Series are averaged within each group, and the risk-factor score is the mean of the three group scores.

2. Benchmark

Long-only, 12-month trend and the plain ECB rate momentum rule, all run through the same account and the same sizing. Window January 2008 to August 2026, 224 months, synthetic 2nd-contract prices. The account starts on 1 January 2008, a month earlier than the earlier reports' 31 January start; that month was a loss, and with the cost change below the frozen spec's one-contract Sharpe goes from +1.09 (earlier reports) to +1.05 here.

Account assumptions:

Item	Setting
Capital	€50,000, euro account, cash earns no interest
Contract	€2,500 per price point, €25 per basis point
Sizing	contracts = round(10% × equity ÷ (σ × €2,500)), set each month-end with the signal
σ	70% trailing 60-day realised volatility + 30% its long-run mean, annualised

Item	Setting
Costs	0.3 bp per side = €7.50 per contract (half a one-tick spread plus exchange, clearing and broker fees); roll = €15 per contract, four times a year
Margin	€237 initial margin per contract (broker quote, September 2026), held flat through history

The 10% volatility target and the volatility blend are my choices, not part of the frozen specification. Section 5 shows what changes with other choices. The 0.3 bp cost is a realistic figure for the ICE contract; 0.5 bp (one full tick) is kept as a sensitivity row and is about what Eurex FEU3 would cost.

3. Variants

17 account runs: 5 strategies at the 10% target (frozen, walk-forward threshold, ECB rate momentum, long-only, 12M trend); the frozen spec at 5%, 15% and 20% targets, with unblended 60-day volatility, with 1 and 3 fixed contracts, at 0.5, 1 and 2 bp per side, from a 2002 start, and on the OIS-spliced price series; ECB rate momentum on the OIS-spliced series. No signal parameter was changed or tuned.

4. Results

Variant	Sharpe	CAGR	Max DD	Final equity	Sharpe vs long-only
Frozen spec, 10% vol target	**+0.85**	**9.0%**	**−20.1%**	**€255,769**	**+0.54**
Walk-forward threshold, 10% vol	+0.80	8.4%	−19.4%	€229,306	+0.49
ECB rate momentum only, 10% vol	+0.66	6.8%	−31.1%	€173,508	+0.36
Long-only, 10% vol (benchmark)	+0.30	2.7%	−52.7%	€83,507	0
12M trend, 10% vol (benchmark)	+0.31	2.9%	−35.8%	€85,590	+0.00
Frozen, 5% vol target	+0.72	3.8%	−10.2%	€100,842	+0.41
Frozen, 15% vol target	+0.80	12.6%	−28.8%	€471,227	+0.50
Frozen, 20% vol target	+0.80	17.1%	−36.4%	€999,523	+0.49
Frozen, 60-day vol only	+0.67	8.4%	−26.8%	€230,907	+0.37
Frozen, fixed 1 contract	+1.00	2.8%	−6.5%	€84,408	+0.70
Frozen, fixed 3 contracts	+0.85	6.1%	−18.2%	€153,223	+0.55
Frozen, cost 0.5 bp/side	+0.79	8.3%	−20.1%	€228,110	+0.49
Frozen, cost 1 bp/side	+0.67	6.9%	−20.5%	€177,585	+0.36
Frozen, cost 2 bp/side	+0.43	4.3%	−38.6%	€111,396	+0.13
Frozen, start Jan 2002	+0.78	8.0%	−20.1%	€345,033	n/a
Frozen, OIS-spliced prices 2005–14	+0.86	9.0%	−18.7%	€257,038	n/a
ECB rate momentum, OIS-spliced prices	+0.64	6.5%	−33.1%	€163,595	n/a

Frozen spec, account detail:

Measure	Value
Total P&L	+€205,769 on €50,000
Realised volatility	10.8% a year against the 10% target
Worst drawdown	−20.1% (−€45,443 at its deepest in euros), Jan 2011 to Aug 2011, recovered Apr 2012
Longest time under water	980 trading days
Worst month / best month	−13.5% (Mar 2026) / +13.2% (Jul 2012)
Winning months	64%
Contracts held	average 15.4, range 1 to 33 as equity grew and volatility fell
Costs paid	€22,612, about 10% of gross profit
Direction changes	12 in 18.7 years; position resized in 151 of 224 months

Margin deployed (Figure 7, bottom panel):

Period	Avg contracts	Avg margin	Share of equity	Peak share if margin had scaled with volatility
2008–12	7.0	€1,652	1.7%	4.5%
2013–21	20.1	€4,768	3.3%	4.8%

Period	Avg contracts	Avg margin	Share of equity	Peak share if margin had scaled with volatility
2022–26	15.5	€3,682	1.6%	3.5%
Whole window	15.4	€3,661	2.4% (peak 4.8%)	4.8%

Calendar years, frozen spec:

Year	P&L €	Return	Year-end equity	Avg contracts (+ long, − short)
2008	+7,647	+15.3%	57,647	−2.0
2009	+32,334	+56.1%	89,980	+4.9
2010	+18,442	+20.5%	108,422	+8.0
2011	−13,859	−12.8%	94,563	−0.2
2012	+44,203	+46.7%	138,766	+12.1
2013	−10,084	−7.3%	128,682	+15.8
2014	+8,304	+6.5%	136,986	+19.7
2015	+1,191	+0.9%	138,178	+16.6
2016	+1,687	+1.2%	139,865	+21.7
2017	+11,763	+8.4%	151,628	+14.9
2018	−1,421	−0.9%	150,207	+23.1
2019	−1,087	−0.7%	149,120	+22.1
2020	+502	+0.3%	149,622	+17.9
2021	+2,787	+1.9%	152,409	−0.2
2022	+41,928	+27.5%	194,338	−11.6
2023	+7,144	+3.7%	201,481	−8.3
2024	+62,057	+30.8%	263,538	+1.3
2025	+1,498	+0.6%	265,036	+20.1
2026 (to Aug)	−9,267	−3.5%	255,769	+12.3

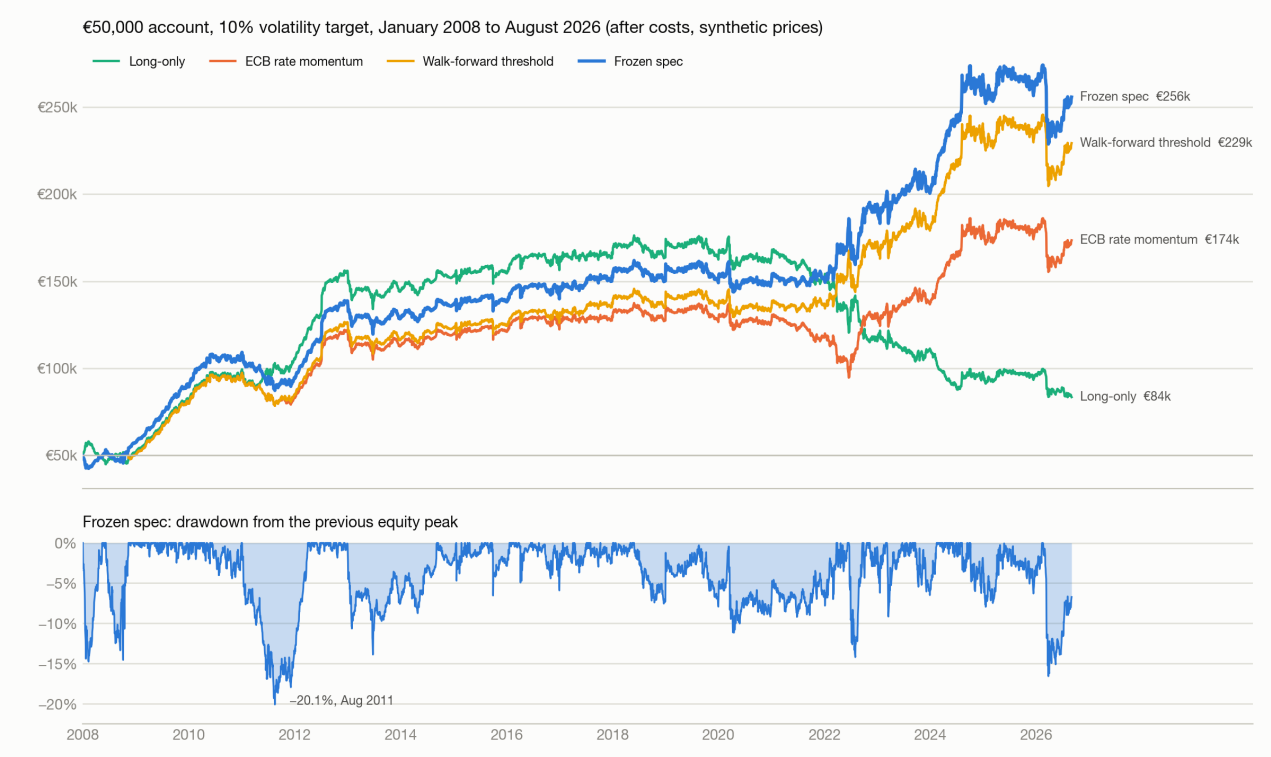


Figure 6. Equity and drawdown, January 2008 to August 2026

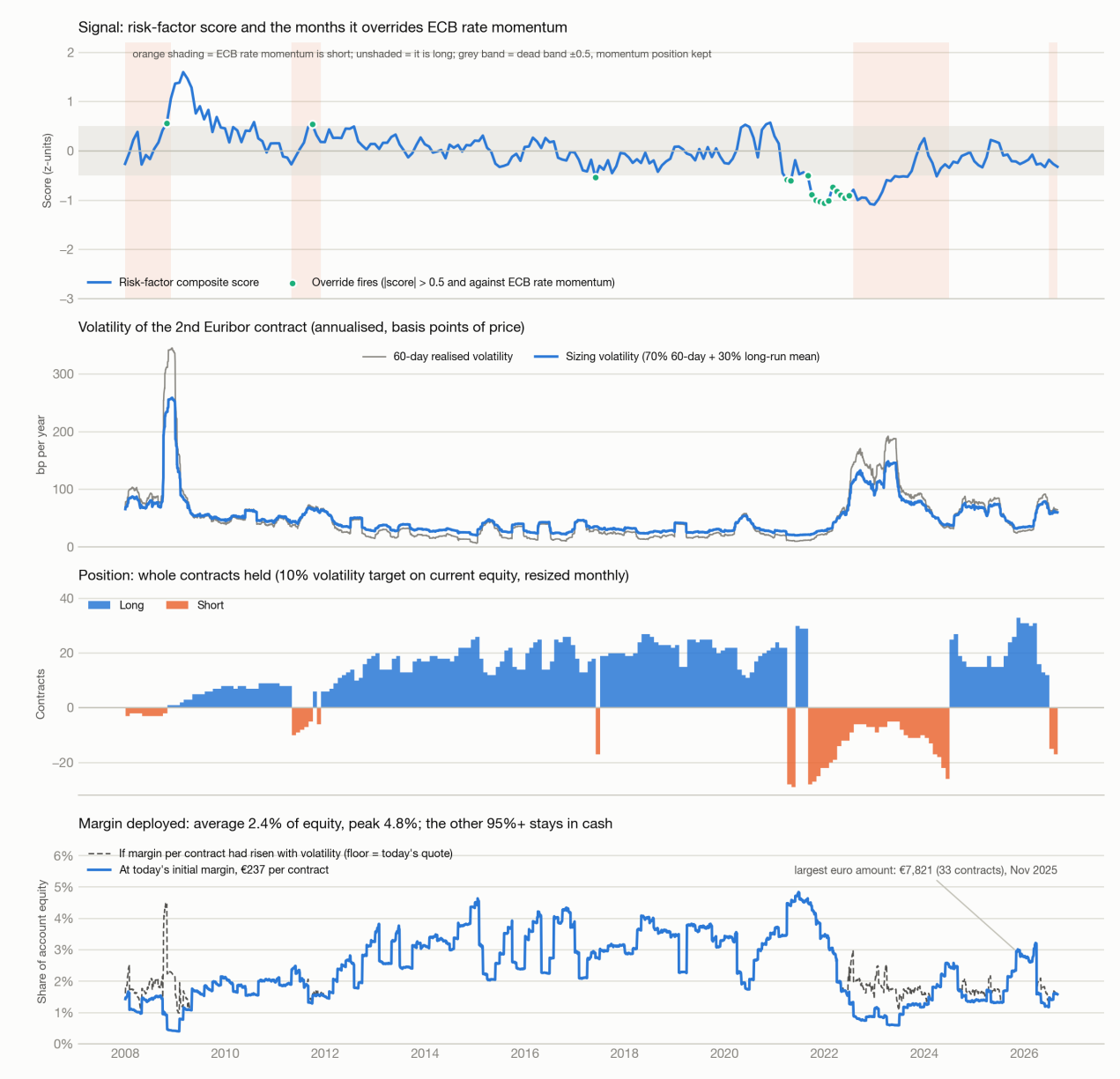


Figure 7. Signal, volatility, contracts and margin

5. Robustness

Regimes. The result is regime dependent, not universal.

Sharpe by period	2008–12 crisis	2013–19 ZLB/QE	2020–21 COVID	2022–26 hikes + cuts
Frozen spec	+1.71	+0.16	+0.17	+0.92
Walk-forward threshold	+1.56	+0.14	+0.16	+0.91
ECB rate momentum	+1.50	+0.10	−0.37	+0.70
Long-only	+1.96	+0.11	−0.37	−0.90

63% of the account's growth came in 2008–12, 32% in 2022–26 and 6% in the nine years 2013–21. The best 5 months are 36% of total P&L and the best 10 are 59%.

Monte Carlo, block bootstrap (12-month blocks, 10,000 paths of 224 months, account returns):

	Frozen spec	Walk-forward	ECB rate momentum	Long-only
Sharpe, 5th / 50th / 95th percentile	+0.38 / +0.86 / +1.39	+0.36 / +0.80 / +1.28	+0.19 / +0.65 / +1.16	−0.24 / +0.28 / +0.89
P(Sharpe > 0)	100%	100%	99%	81%
P(beats long-only) / P(beats ECB rate momentum)	97% / 97%	96% / 93%		
Final equity, 5th / 50th / 95th	€95k / €249k / €746k	€90k / €224k / €623k	€65k / €169k / €504k	€27k / €80k / €284k
P(ending below €50,000)	0.2%	0.3%	1.7%	25%
Max drawdown, median / 5th / 1st percentile	−19% / −32% / −39%	−19% / −32% / −39%	−24% / −40% / −48%	−38% / −63% / −72%
P(drawdown worse than 20% / 30%)	44% / 8%	44% / 7%	71% / 23%	96% / 74%
First-year P&L, 5th / 50th / 95th	−€5.4k / +€2.7k / +€25.7k	−€5.1k / +€2.4k / +€25.4k	−€6.0k / +€1.9k / +€23.6k	−€9.8k / +€0.3k / +€26.1k
P(loss after 1 year / 5 years)	25% / 8%	25% / 8%	34% / 15%	47% / 39%

No-skill nulls (the long/short path shifted or block-shuffled in time, position sizes left on their dates):

	Observed Sharpe	Null mean	Null 95th pct	p, circular shift	p, block permutation
Frozen spec	+0.86	+0.09	+0.51	0 of 212	0 of 3,000
Walk-forward threshold	+0.80	+0.10	+0.52	0.005	0.001
ECB rate momentum	+0.66	+0.14	+0.51	0.042	0.007

Deflated Sharpe (expected best-of-N Sharpe under the null: 0.53 for 45 trials, 0.57 for 70): frozen spec 0.91 / 0.88, walk-forward threshold 0.86 / 0.82, ECB rate momentum 0.70 / 0.65, long-only 0.17 / 0.13. Monthly returns have skew −0.39 and kurtosis 7.0.

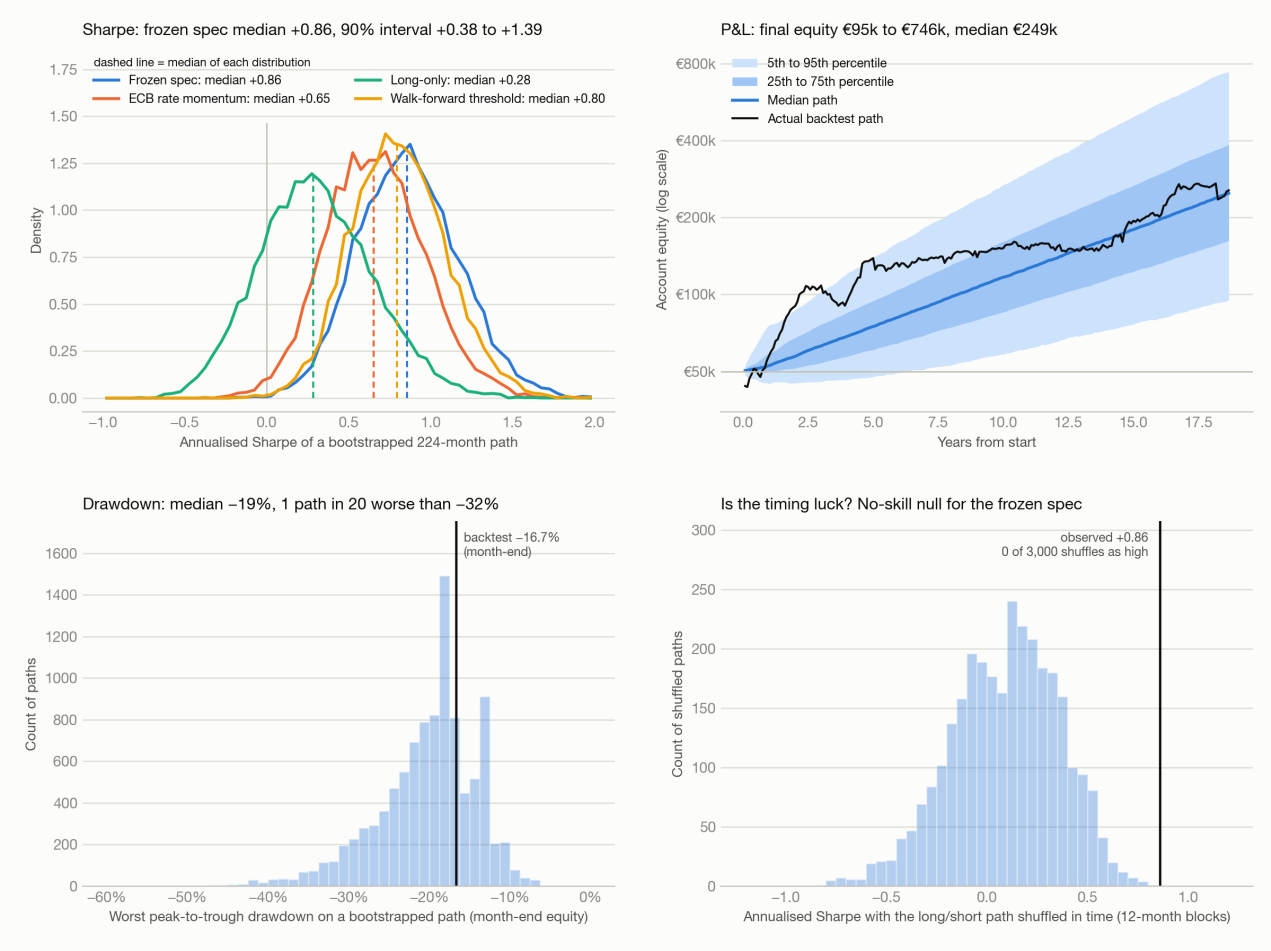


Figure 8. Monte Carlo, full window

Sizing. Volatility targeting costs Sharpe here: +1.00 with one fixed contract against +0.85 at a 10% target, because the strategy earns most when volatility is high, which is when the target cuts the position. Without the long-run blend the account ran up to 73 contracts in the zero-rate years and Sharpe fell to +0.67. Higher targets scale return and drawdown together (20% target: −36.4% drawdown, a −27% month).

Costs. Sharpe +0.79 at 0.5 bp per side (one full tick), +0.67 at 1 bp and +0.43 at 2 bp. Rolls are 76% of the contract-sides traded: at a 10% target the account rolls about 15 contracts four times a year, against 12 direction changes in 18.7 years.

Instrument. On the OIS-spliced prices for 2005–14 the frozen spec is +0.86 (€257k) against +0.85 (€256k); ECB rate momentum goes from +0.66 to +0.64.

Margin. Margin is never the constraint. It averaged 2.4% of equity and peaked at 4.8%; even with margin scaled up with volatility the peak stays at 4.8%. The binding limit is the volatility target, and 95% of the account sits in cash, which this backtest credits no interest on.